

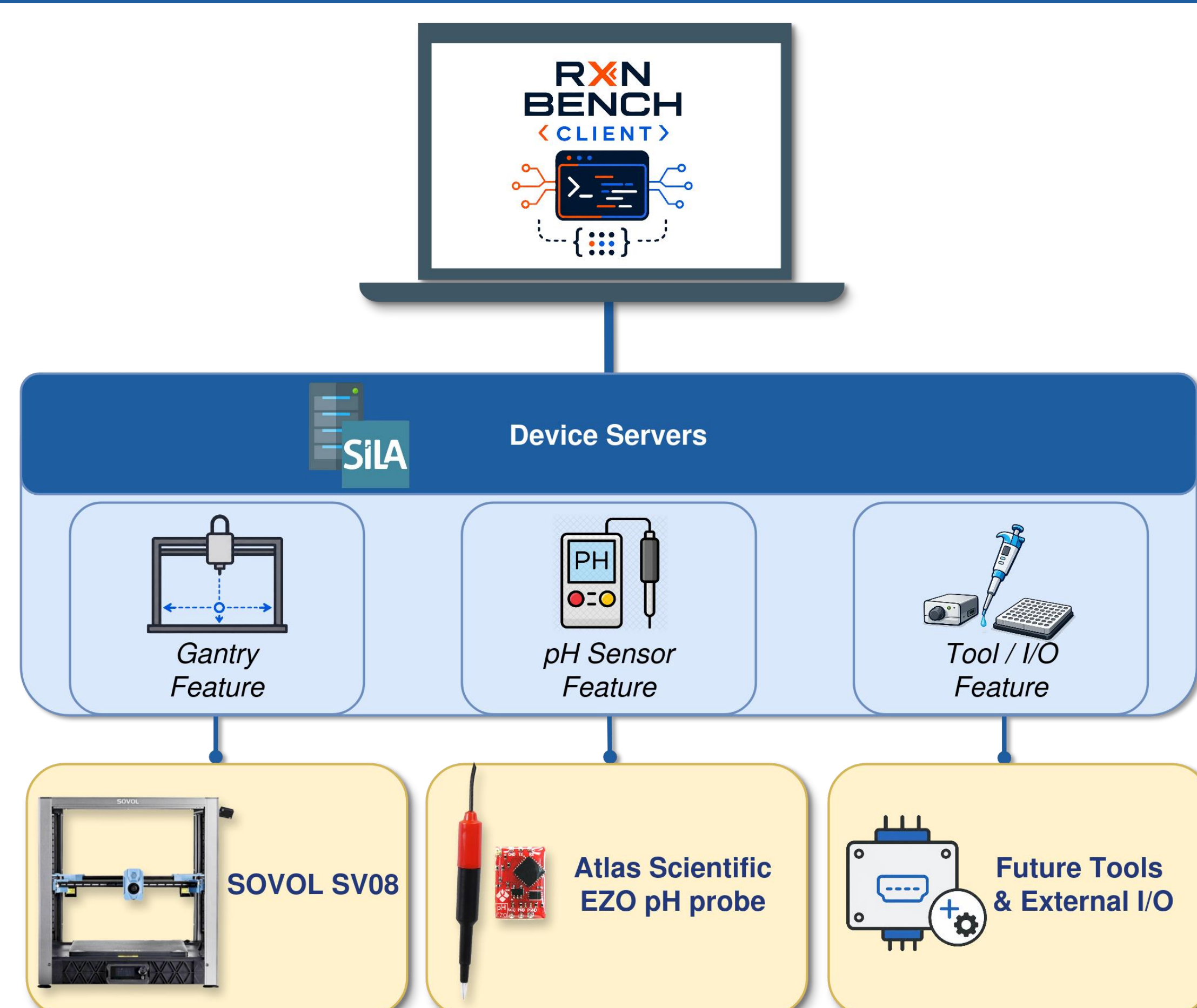
Introduction

- Discovery increasingly runs on data.** Modern science depends on it; machine learning, high-throughput screening, and AI-assisted discovery all require large, reliable experimental datasets. Generating that data by hand is slow, inconsistent, and labor-intensive, making automated data collection a growing bottleneck for research.
- Automation is the answer - but it's expensive and rigid.** Automated platforms improve repeatability, cut manual workload, and generate data at scale, but most remain costly, specialized, and locked to specific hardware and workflows. This puts automated experimentation out of reach for many labs.
- Rxn Bench: accessible automation.** A low-cost, modular platform combining accessible hardware, modern lab software frameworks, and customizable 3D-printable components - letting labs integrate diverse devices, adapt workflows, and generate data at scale for under \$1000.

Design Goals

Low-Cost Modular Hardware Independent Programmable Expandable

System Design

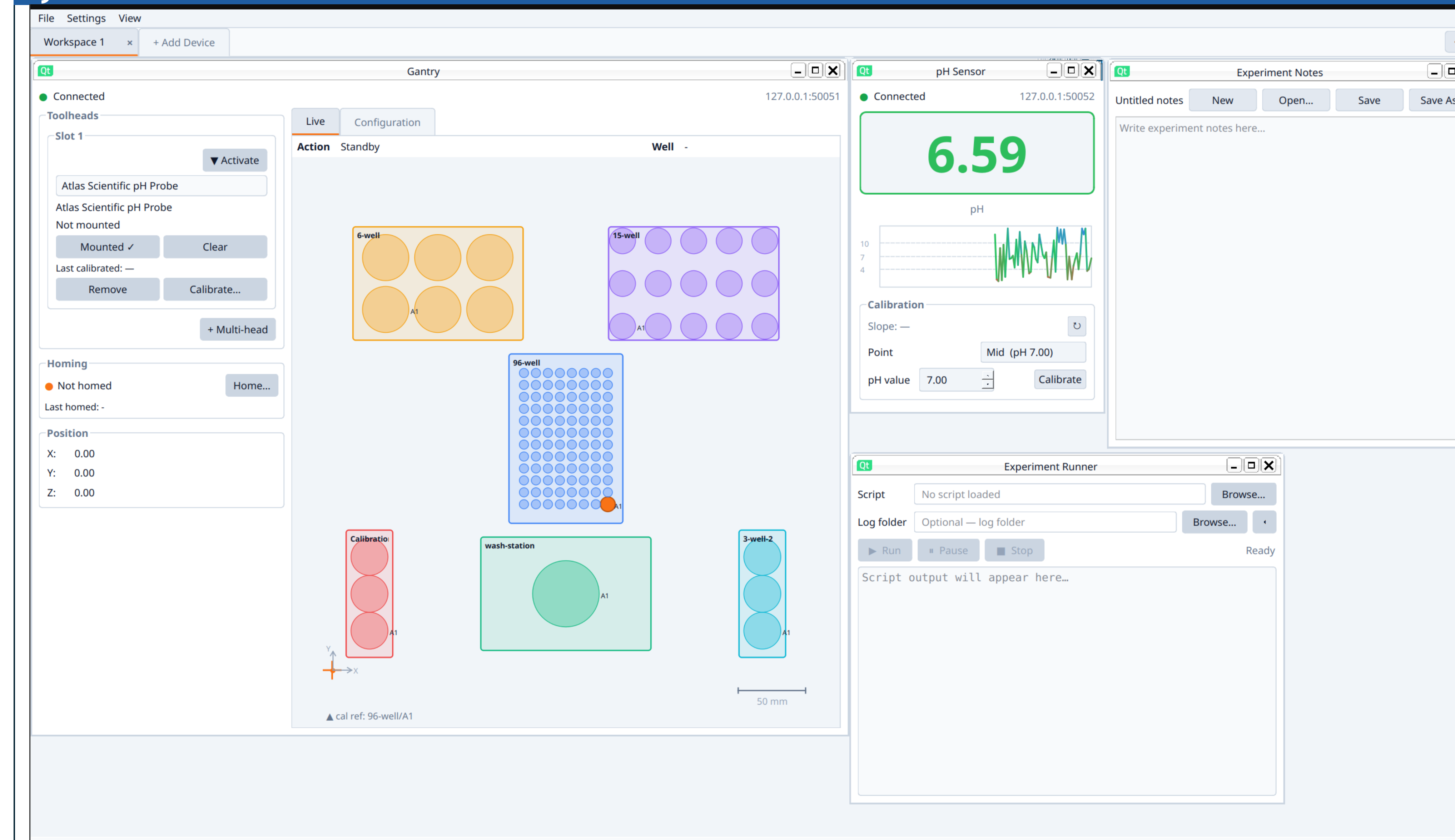


• Rxn Bench is a modular lab-automation platform for programmable scientific workflows, built from a repurposed SOVOL SV08 3D printer with custom 3D-printed components and open software interfaces.

• **Capability-based hardware abstraction.** Devices are represented through [SiLA2](#) feature servers running on a backend computer (e.g., a Raspberry Pi), which expose standardized commands and properties for each capability: pH sensor, gantry, pump, etc. New hardware plugs into the same feature interface, so a device can be swapped or added with minimal frontend and backend changes. *If it walks like a duck and talks like a duck, it's treated as a duck.*

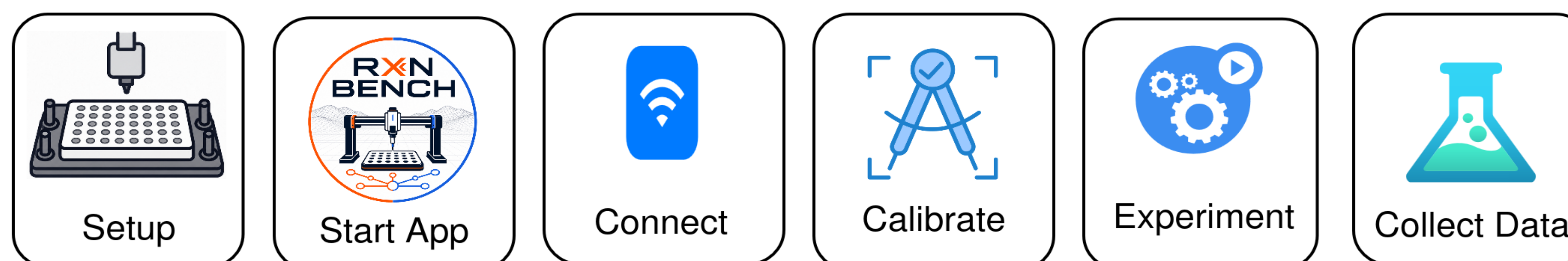
• **Desktop control + Python API.** Rxn Bench Client provides a desktop interface for configuration and direct device control, while a Python API enables headless operation and experiment scripting.

System Interface

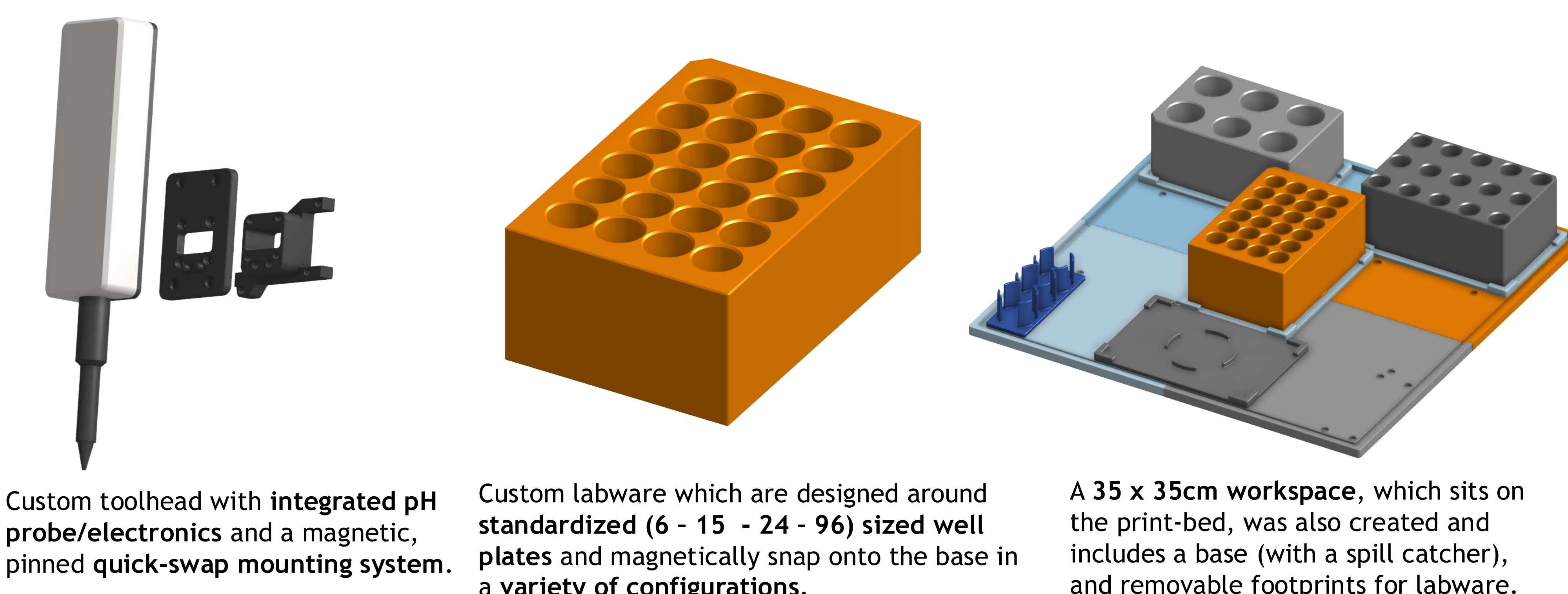


Rxn Bench desktop user interface for device control and executing automated workflows. The workspace integrates device-specific widgets, including XYZ gantry control (left), pH sensor monitoring (right), and a script-based experiment runner (bottom right).

System Workflow



Custom 3D Printed Labware Tools and Mounts

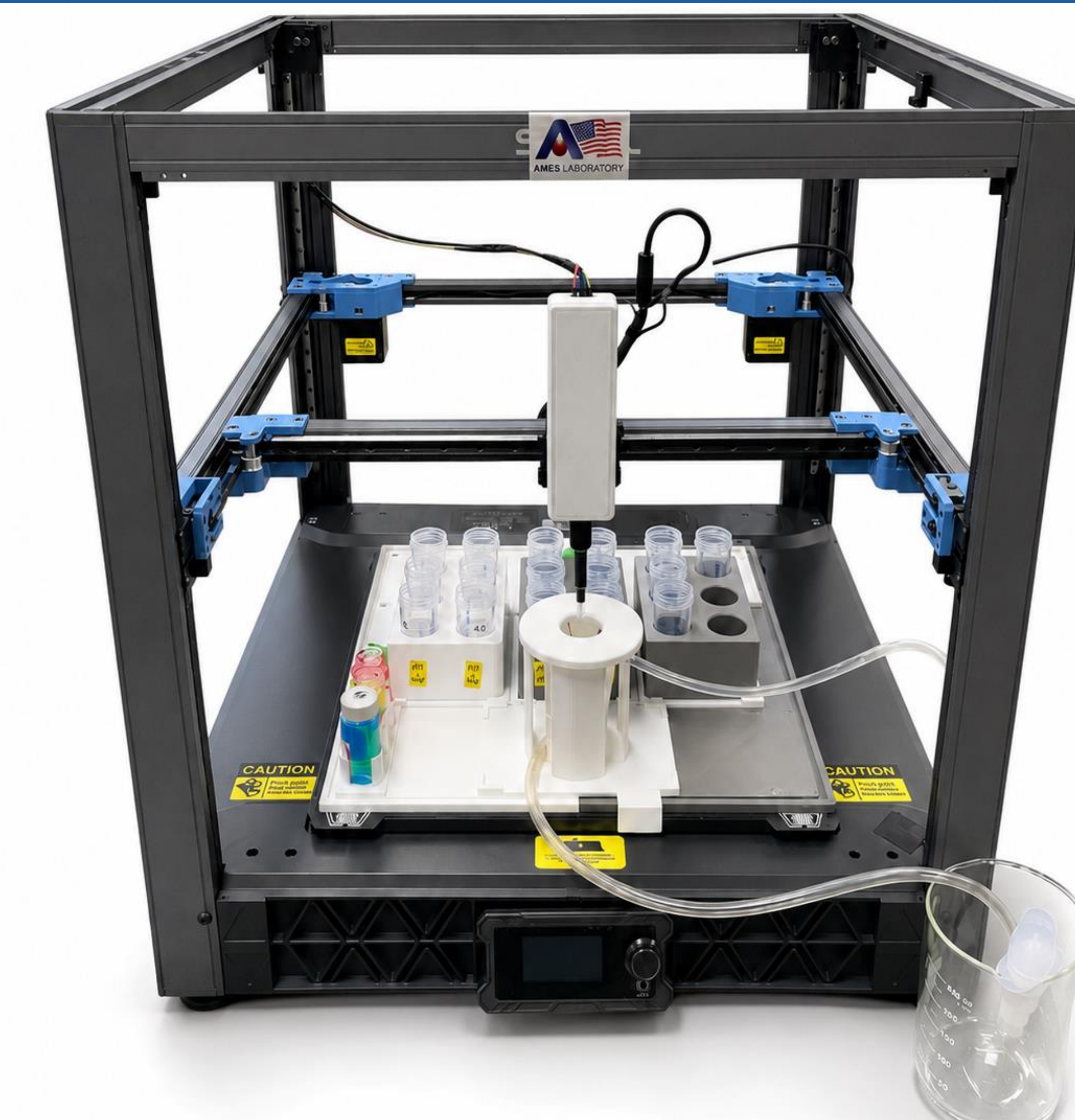


Custom toolhead with integrated pH probe/electronics and a magnetic, pinned quick-swap mounting system.

Custom labware which are designed around standardized (6 - 15 - 24 - 96) sized well plates and magnetically snap onto the base in a variety of configurations.

A 35 x 35cm workspace, which sits on the print-bed, was also created and includes a base (with a spill catcher), and removable footprints for labware.

Prototype Platform



Rxn Bench setup for performing automated pH measurements, including a pH toolhead, calibration station (left), sample plates (middle), and a washing station (bottom) to wash the probe tip between measurements.



WATCH RXN BENCH IN ACTION

Scan for a short demonstration of Rxn Bench performing automatic pH calibration and measurements

<https://www.youtube.com/shorts/6GllqKrOk-o>

Performance, Capabilities and Future Plans

- Rxn Bench automated two- and three-point pH-probe calibration and collected measurements from more than 24 samples without requiring continuous manual operation.
- Demonstrated capabilities**
 - Automated two- and three-point pH-probe calibration
 - Automated measurement of more than 24 samples
 - Configurable workspaces for multiple labware formats
 - Swappable toolheads and capability-based device integration
 - Programmable experiment workflows through a Python client
 - Automation below \$1000
- Future development**
 - Multi-toolhead operation
 - Automated liquid handling
 - Closed-loop and self-driving experiments
 - AI-assisted device integration and experiment optimization